

The Module Structure of Integral Designs*

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The existence problem for t -designs with prescribed parameters is solved by allowing positive and negative integral multiplicities for the blocks.

INTRODUCTION

In this paper we discuss design of type $(1, t)$, in the sense of Dembowski [1], or what are often called t -designs. Repetitions of blocks are permitted and will be described by associating to each block non-negative integral multiplicities. Superposition of designs means addition of associated multiplicities and is a natural addition for designs. The module structure arising by also allowing negative integral multiplicities will be completely determined. Moreover, we will be able to characterize completely the parameter values which can occur with integral designs. In contrast, the corresponding problem for designs with non-negative multiplicities is one of the main unsolved problems of design theory. In the previous paper [2], we gave up the requirement of constant block size in order to define several multiplications for general designs; the corresponding algebra structures are generally lost now, but they can still be used in the constructions of special designs.

0.1. Let $t, v, k, \lambda_0, \lambda_1, \dots, \lambda_t$ all the integers where $v \geq 1$ and $0 \leq t, k \leq v$. Let $V = \{1, 2, \dots, v\}$, $X = \{x \mid x \subseteq V\}$, and $V_k = \{x \mid x \subseteq V, |x| = k\}$. The members of X are called blocks, those in V_k are called k -blocks or blocks of size k . An *integral* $(v, k, \lambda_0, \dots, \lambda_t)$ -*design* associates integral multiplicities $c(x)$ to k -blocks x and zero to all other blocks such that:

$$\ell(x) = \sum_{y \supseteq x} c(y) = \lambda_s, \quad \text{if } |x| = s \leq t.$$

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Our main result may now be stated. Given integers $t, v, k, \lambda_0, \dots, \lambda_t$ as above, there exists an integral $(v, k, \lambda_0, \dots, \lambda_t)$ -design if and only if

$$\lambda_{s+1} = \frac{k-s}{v-s} \lambda_s, \quad \text{for } 0 \leq s < t. \quad (0.1)$$

In proving this result, we develop methods for constructing a design with prescribed parameters $\lambda_0, \dots, \lambda_t$ satisfying (0.1). Any other design with the same parameters may be obtained by adding an arbitrary integral design with all parameters zero; such a design is called a *null design*.

The problem of possible parameters where the constant block size restriction is dropped was solved in [2]; in that case there is an integral design with any given collection of integral parameters $\lambda_0, \dots, \lambda_t$.

0.2. Fixing t, v , and k , the set of all integral $(v, k, \lambda_0, \dots, \lambda_t)$ -designs form a module $C_{tk}(v)$ over Z and the null designs form a submodule $N_{tk}(v)$. Both are contained in the module Z_+^X of integral valued functions on X . It is well known that the structure of a submodule of Z_+^X is completely determined by the dimension of the submodule. The dimensions of $C_{tk}(v)$ and $N_{tk}(v)$ are computed in Section 2. Furthermore we prove the existence of a module basis for $N_{tk}(v)$ satisfying a minimality condition (Section 4), this basis is then used in an essential way in the construction of integral designs with given parameters (Section 5).

0.3. The transform $\hat{c} = Gc$, defined in general by:

$$\hat{c}(u) = (Gc)(u) = \sum_{v \supseteq u} c(y), \quad (0.2)$$

was recognized in our preceding paper to be some sort of Fourier transform. By partitioning X into $V_0, V_1, \dots, V_v$ the vectors c and $d = \hat{c}$ split into blocks c_k (the restriction of c to V_k) and d_t (the restriction of d to V_t); furthermore the matrix G or $G(v)$ splits into blocks G_{tk} or $G_{tk}(v)$ such that:

$$d_t = \sum_{k=0}^v G_{tk} c_k, \quad \text{for } 0 \leq t \leq v. \quad (0.3)$$

The properties of G_{tk} are, of course, most relevant for design theory; in fact, a necessary and sufficient condition that c , of constant block size $k \geq t$, belong to C_{tk} is that

$$G_{tk} c_k = \lambda_t e_t, \quad (0.4)$$

where e_t is the vector which has components one for each t block. Sometimes it will be convenient to let e_t also denote the design which has multi-

plicity one for each t block and zero for all other blocks. (0.4) is one form of a fundamental theorem of design theory; it is proved in Section 1 as a special case of a basic matrix identity for the G_{tk} . Our main result then yields necessary and sufficient conditions under which (0.4) has an "integral solution."

There exists an integral c_k satisfying (0.4) if and only if there exist integers $\lambda_0, \dots, \lambda_{t-1}$ which, along with λ_t satisfy the conditions (0.1).

1. A FUNDAMENTAL MATRIX IDENTITY

This section is devoted to the ramifications of the matrix equation:

$$\binom{k-s}{h-s} G_{sk} = G_{sh} G_{hk}, \quad \text{where} \quad s \leq h \leq k. \quad (1.1)$$

1.1. Before we prove (1.1), let us consider more closely the structure of G_{tk} . If we define $[u, x]$ to be one if $u \subseteq x$ and zero otherwise, we have

$$G = [[u, x]]_{2^v \times 2^v}.$$

The block G_{tk} is then the $\binom{v}{t} \times \binom{v}{k}$ inclusion matrix of the t -blocks with the k -blocks:

$$G_{tk} = [[u, x]], \quad \text{where} \quad u \in V_t \quad \text{and} \quad x \in V_k. \quad (1.2)$$

We may now prove (1.1) by observing that if $|u| = s$ and $|z| = k$ then

$$\sum_{|x|=h} [u, x][x, z] = \binom{k-s}{h-s} [u, z].$$

If c has constant block size k , then we may identify c with the block c_k in its decomposition and in this special case (0.3) becomes:

$$(\hat{c})_s = G_{sk} c_k, \quad \text{where } t \text{ is replaced by } s.$$

Combining this with (1.1) yields:

$$(\hat{c})_s = \frac{1}{\binom{k-s}{h-s}} G_{sh} (\hat{c})_h, \quad \text{where} \quad s \leq h \leq k. \quad (1.3)$$

1.2. FUNDAMENTAL THEOREM. Let $0 \leq t \leq k \leq v$. If $c \in Z_+^x$ has constant block size k and if $(\hat{c})_t = \lambda_t e_t$, then c is an integral $(v, k, \lambda_0, \dots, \lambda_t)$ -design.

Furthermore

$$\lambda_s = \frac{\binom{v-s}{t-s}}{\binom{k-s}{t-s}} \lambda_t, \quad \text{for } s = 0, 1, \dots, t.$$

We observe that

$$G_{st}e_t = \binom{v-s}{t-s} e_s.$$

Substituting $(\hat{c})_t = \lambda_t e_t$ in (1.3) (where h is replaced by t) yields

$$(\hat{c})_s = \frac{1}{\binom{k-s}{t-s}} G_{st} \lambda_t e_t = \lambda_t \frac{\binom{v-s}{t-s}}{\binom{k-s}{t-s}} e_s.$$

This may also be proved by a straightforward counting argument (see [1]).

COROLLARY. *An integral $(v, k, \lambda_0, \dots, \lambda_t)$ design, where $t \leq k$, is null if and only if $\lambda_i = 0$ for some $i \leq t$.*

1.3. THEOREM. *Let $0 \leq t \leq h \leq k \leq v$. If $c \in C_{tk}$ (or N_{tk}) then $d = (\hat{c})_h \in C_{th}$ (or N_{th}). Furthermore if c is non-negative then d is non-negative with multiplicities less than or equal to $\lambda_t(c)$, (where $\lambda_0(c), \dots, \lambda_t(c)$ are the parameters of c).*

We have:

$$(\hat{d})_t = G_{th}d_h = G_{th}(\hat{c})_h = G_{th}G_{hk}c_k = \binom{k-t}{h-t} G_{tk}c_k = \binom{k-t}{h-t} \lambda_t(c) e_t.$$

Now, if $u \subseteq x$ where $|u| = t$ and $|x| = h$, then u is contained in every k -block containing x . The multiplicity of x is the sum of the multiplicities of the k -blocks containing it, but since these all contain u their sum cannot exceed $\lambda_t(c)$.

1.4. In [2] we computed the inverse of G :

$$G^{-1} = [(-1)^{|u|+|x|} [u, x]].$$

With respect to the natural partition of X , we have the decomposition:

$$G^{-1} = [(-1)^{t+k} G_{tk}].$$

Using this, we may prove the sufficiency of conditions (1.3) in order that a cover have constant block size.

THEOREM. *Let $c \in Z_+^X$. In order that c has constant block size k it is necessary and sufficient that*

$$(\hat{c})_s = \frac{1}{k-s} G_{s,s+1}(\hat{c})_{s+1}, \quad \text{if } s < k, \quad (1.4)$$

and

$$(\hat{c})_s = 0 \text{ if } s > k. \quad (1.5)$$

The necessity of the conditions (1.4) follows since they are included among the conditions (1.3), and the necessity of the conditions (1.5) follows directly from the definition of $\hat{c}$, (0.2). One should note at this time that conditions (1.4) imply the full set of conditions (1.3). Now assume that $\hat{c}$ satisfies (1.4) and (1.5). We have

$$c_h = (G^{-1}\hat{c})_h = \sum_s (-1)^{|s|+|h|} G_{hs}(\hat{c})_s.$$

It follows from (1.2) that G_{hs} is the zero matrix if $h > s$; it follows from this and (1.5) that

$$c_h = \sum_{s=h}^k (-1)^{|s|+|h|} G_{hs}(\hat{c})_s, \quad \text{for } h \leq k,$$

and

$$c_h = 0, \quad \text{for } h > k.$$

Using (1.3) with $h = k$ to evaluate $(\hat{c})_s$ yields

$$c_h = \sum_{s=h}^k (-1)^{|s|+|h|} G_{hs} G_{sk}(\hat{c})_k,$$

which in turn gives

$$c_h = (-1)^{|h|} \sum_{s=h}^k (-1)^{|s|} \binom{k-h}{s-h} G_{hk}(\hat{c})_k.$$

Thus $c_h = 0$ if $h < k$. Finally we observe that G_{kk} is the identity, hence $c_k = (\hat{c})_k$.

One result of this theorem is that the conditions that c be a $(v, k, \lambda_0, \dots, \lambda_t)$ -design may now be stated entirely in terms of the transform G .

THEOREM. Let $0 \leq t \leq k \leq v$ and let

$$\lambda_{s+1} = \frac{k-s}{v-s} \lambda_s, \quad \text{for } 0 \leq s < t.$$

Then $c \in Z_+^x$ is an integral $(v, k, \lambda_0, \dots, \lambda_t)$ -design if and only if

$$(\hat{c})_h = \lambda_h e_h, \quad \text{for } 0 \leq h \leq t,$$

$$(\hat{c})_h = \frac{1}{k-h} G_{h,h+1} \hat{c}_{h+1}, \quad \text{for } t \leq h < k,$$

and

$$(\hat{c})_h = 0, \quad \text{for } k < h \leq v.$$

2. THE STRUCTURE OF $G_{tk}(v)$

As we have noted, the structure of G_{tk} is obvious for some values t and k :

$$G_{tk} = 0, \quad \text{if } k < t,$$

$$G_{tt} = I,$$

$$G_{0t} = J,$$

and

$$G_{tv} = J,$$

where 0 is the zero matrix, I the identity matrix, and J the matrix of all ones, all of appropriate dimensions.

2.1. Using the above special cases, the structure of $G_{tk}(v)$ is given inductively by:

$$G_{tk}(v) = \left[\begin{array}{c|c} G_{tk}(v-1) & G_{t,k-1}(v-1) \\ \hline 0 & G_{t-1,k-1}(v-1) \end{array} \right], \quad \text{where } 1 \leq t, \quad k \leq (v-1).$$

This block decomposition is obtained by partitioning V_k and V_t each into two classes: those blocks avoiding the last point v in one class and those containing v in the other. The upper left-hand block is then the inclusion matrix of those t -blocks avoiding v with those k -blocks avoiding v , which is $G_{tk}(v-1)$. The lower left-hand block is the inclusion matrix of the t -blocks containing v with the k -blocks avoiding v and hence is zero. The upper right-hand block is the inclusion matrix of the t -blocks avoiding v with the k -blocks containing v , which is the inclusion matrix

of the t -blocks of $\{1, \dots, v-1\}$ with the $k-1$ blocks of $\{1, \dots, v-1\}$, i.e., the upper right-hand block is $G_{t, k-1}(v-1)$. Similarly the lower right-hand block is $G_{t-1, k-1}(v-1)$.

There are other structures theorems for the $G_{tk}(v)$. A rather extensive study of the matrices $G_{t-1, t}(v)$ occurs in [3]; in the terminology of that paper, $G_{t-1, t}(v) = P_t(1, 1, \dots, 1)$.

2.3. This paragraph is devoted to showing that $G_{tk}(v)$ is non-singular in the case in which $v = t + k$ and $t \leq k$. Let u be a fixed t -block and partition both V_k and V_t into $t+1$ cells where the t - (or k)-blocks which intersect u in exactly i points form the i -th cell. This yields the block decomposition:

$$G_{tk} = [A_{ij}], \quad \text{where } 0 \leq i, j \leq t.$$

The matrix A_{ij} is the inclusion matrix of those t -blocks which intersect u in i -points with those k -blocks which intersect u in j -points. It is easy to verify that the row sums of A_{ij} are a constant a_{ij} . We observe that $a_{ij} = 0$ if $i > j$ and that

$$a_{ii} = \binom{k-t+i}{k-t} \neq 0.$$

Now consider the $(t+1) \times (t+1)$ matrix $[a_{ij}]$. We have just noted that it is triangular and non-singular. We may therefore find a column vector γ such that:

$$\gamma = \begin{bmatrix} \gamma_0 \\ \vdots \\ \gamma_1 \end{bmatrix} \text{ satisfies } [a_{ij}]\gamma = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

If we define $c(x) = \gamma_j$, where $|x \cap u| = j$ and $|x| = k$, and $c(x) = 0$ otherwise, then $G_{tk}c_k = \delta_u$, where $\delta_u(x) = 0$ if $x \neq u$ and $\delta_u(u) = 1$. Since u was arbitrary, the linear transformation G_{tk} is onto and hence non-singular.

2.4. We will now prove that in general $G_{tk}(v)$ is of full rank when $0 \leq t \leq k \leq v$.

THEOREM.

$$\text{rank } G_{tk}(v) = \binom{v}{t}, \quad \text{when } t \leq k \leq v-t,$$

$$\text{rank } G_{tk}(v) = \binom{v}{k}, \quad \text{when } v-t \leq k \leq v, \quad t \leq k.$$

The first equation follows immediately from the matrix equation (1.1) and the fact that $G_{t, v-t}(v)$ is non-singular:

$$\binom{v-2t}{k-t} G_{t, v-t}(v) = G_{tk}(v) G_{k, v-t}(v).$$

Since $G_{t, v-t}(v)$ has full row rank, $G_{tk}(v)$ has full row rank. Similarly for $v-t \leq k \leq v$:

$$\binom{2k-v}{k+t-v} G_{v-k, k}(v) = G_{v-k, t}(v) G_{tk}(v).$$

Since $G_{v-k, k}(v)$ has full column rank, $G_{tk}(v)$ has full column rank.

2.5. We define the integer $n_{tk}(v)$ to be

$$\binom{v}{k} - \binom{v}{t}, \quad \text{for } t \leq k \leq v-t,$$

and zero otherwise. Since $G_{tk}(v)$ is a $\binom{v}{t}$ by $\binom{v}{k}$ matrix of full rank when $t \leq k$, its null space has dimension $n_{tk}(v)$. One result of the fundamental theorem, paragraph 1.2, is that $N_{tk}(v)$ may be identified with the integral vectors in the null space of $G_{tk}(v)$, when $t \leq k$. Also it is clear that, when $k < t$, then $N_{tk} = \{0\}$. As a corollary to the theorem of the previous paragraph we may state:

COROLLARY 1. *Dimension $N_{tk}(v) = n_{tk}(v)$ and $N_{tk}(v) \cong Z_+^m$, where $m = n_{tk}(v)$.*

It also follows from the fundamental theorem that $C_{tk}(v)$ has at most one more degree of freedom than $N_{tk}(v)$. However, it is clear that $e_k \in C_{tk}(v)$ for all t , hence $C_{tk}(v) \neq N_{tk}(v)$.

COROLLARY 2. *Dimension $C_{tk}(v) = n_{tk}(v) + 1$ and $C_{tk}(v) \cong Z_+^m$, when $m = n_{tk}(v) + 1$.*

3. FOUNDATIONS, EXTENSIONS, AND SUSPENSIONS

The support of an element $c \in Z^X$ is defined to be the collection of elements of X which have non-zero multiplicities:

$$\text{supp}(c) = \{x \mid c(x) \neq 0\}.$$

The foundation of an element $c \in Z^X$ is defined to be the union of the blocks in its support:

$$\text{found}(c) = \bigcup_{c(x) \neq 0} x.$$

3.1. If c is an integral $(v, k, \lambda_0, \dots, \lambda_t)$ -design with $t, k \geq 1$ which is not null, it is easy to see that $\text{found}(c) = V(\lambda_1 \neq 0)$; hence each point of V is contained in some block x for which $c(x) \neq 0$. If on the other hand $c \in N_{tk}(v)$, it is not immediately clear what, if anything, can be said about its foundation.

Let $c \in N_{tk}$ where $c \neq 0$. We may assume without loss of generality that $\text{found}(c) = \{1, 2, \dots, w\} = W$, where $w \leq v$. It is not difficult to see that c is in a natural way a null t -design on W and hence an element of $N_{tk}(w)$. It follows that $n_{tk}(w)$, the dimension of $N_{tk}(w)$, is positive. That is possible only if $t < k < w - t$. Hence we conclude that $|\text{found}(c)| = w \geq k + t + 1$. The remaining paragraphs in this section are devoted to constructing null t -designs with foundation size $k + t + 1$.

Before closing this paragraph, we would like to make one further observation. We have noted that an element of $N_{tk}(v)$ with foundation in W may naturally be considered an element of $N_{tk}(w)$. The converse is also true. Any element of Z_+^Y , where $Y = \{x \mid x \subseteq W\}$, may be identified with its extension by zero to an element in Z_+^X ; it is not difficult to see that a null t -design on W is thereby identified with a null t -design on V . Thus in a natural way $N_{tk}(w)$ is a submodule of $N_{tk}(v)$.

3.2. In [2] we defined a convolution $*$ as follows:

$$(c * d)(z) = \sum_{x+y=z} c(x) d(y) \quad (\text{where } + \text{ is Boolean sum}). \quad (3.1)$$

One of the results of that previous paper is that if c and d are t -designs then $c * d$ will satisfy all of the conditions of a t -design except the requirement of constant block size; generally this requirement will not be satisfied. A sufficient condition under which $c \in C_{t,k_1}(v)$, $d \in C_{t,k_2}(v)$ implies that $c * d \in C_{t,k_3}(v)$ is that $|x \cap y|$ is fixed whenever $c(x) d(y) \neq 0$.

One method of ensuring that $|x \cap y|$ is constant whenever $c(x) d(y) \neq 0$ is to choose c and d with disjoint foundations. As we have noted, this is possible only if c and d are null designs.

3.3. In the following we use the Boolean product $xy = x \cap y$. If we assume that $\text{found}(c) = u$ and $\text{found}(d) = w$ are disjoint, then (3.1) becomes

$$(c * d)(z) = \begin{cases} c(uz) d(wz), & \text{if } z \subseteq u \cup w, \\ 0, & \text{otherwise.} \end{cases}$$

Continuing under this assumption, we have

$$\widehat{c * d}(z) = \begin{cases} \hat{c}(uz) \hat{d}(wz), & \text{if } z \subseteq u \cup w, \\ 0, & \text{otherwise.} \end{cases} \quad (3.2)$$

We prove (3.2) as follows:

$$\begin{aligned} \widehat{c * d}(z) &= \sum_{y \supseteq z} (c * d)(y) \\ &= \sum_{z \subseteq y \subseteq u \cup w} c(uy) d(wy) \\ &= \sum_{zu \subseteq x \subseteq u} c(x) \sum_{zw \subseteq x \subseteq w} d(x) \\ &= \begin{cases} \hat{c}(zu) \hat{d}(zw), & \text{if } z \subseteq u \cup w, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

We should note that (3.2) is a special case of a more general equation considered in [2]:

$$(\widehat{c * d})(z) = \sum_{x \cup y = z} (-2)^{|xy|} \hat{c}(x) \hat{d}(y).$$

At this point we wish to introduce a convention:

$$N_{-1k}(v) = \{c \mid c \in Z^X \text{ and } c(x) = 0 \text{ whenever } |x| \neq k\}.$$

THEOREM. *Let $-1 \leq t \leq k$, $-1 \leq s \leq h$, and $0 \leq h, k \leq v$. If $c \in N_{tk}(v)$ and $d \in N_{sh}(v)$ and if $u = \text{found}(c)$, $w = \text{found}(d)$ are disjoint, then $c * d \in N_{t+s+1, k+h}(v)$.*

If $|z| \leq t + s + 1$, then $|uz| \leq t$ or $|wz| \leq s$ and it follows from (3.2) that $(c * d)(z) = 0$. Furthermore if $(c * d)(z) \neq 0$, then $c(uz)$ and $d(wz)$ are non-zero and $z \subseteq u \cup w$. It follows then from the disjointness of u and w that $|z| = |uz| + |wz| = k + h$.

3.4. In this paragraph we will use the general method of construction in two very simple cases. For $x \in X$ we define $\delta_x \in Z_+^X$ to be the function which has value one at x and zero elsewhere. It is easy to see that $\{\delta_x \mid |x| = k\}$ is a natural modular basis for $N_{-1,k}(v)$. If $x \subseteq V$, $c * \delta_x$ is called the *extension* of c by x . If $c \in N_{t,k}(v)$, $|x| = h$ and x is disjoint from the foundation of c , then, by the main result of the last paragraph, $c * \delta_x \in N_{t,k+h}(v)$.

If p and q are distinct points of V , let $d_{pq} = \delta_{\{p\}} - \delta_{\{q\}}$. Then $c * d_{pq}$

is called the suspension of c by p and q . If $c \in N_{t,k}(v)$ and p, q is disjoint from the foundation of c , then $c * d_{pq} \in N_{t+1,k+1}(v)$. This follows since $d_{pq} \in N_{0,1}(v)$.

Now assume that $t < k < v - t$; this implies that $k + t + 1 \leq v$. Choose u to be any $(k + t + 1)$ -subset of V , label $2t + 2$ of the points in u $p_0, p_1, \dots, p_t, q_0, q_1, \dots, q_t$, and let x be the subset of the $k - t - 1$ points remaining in u . As we have noted $d_{p_i q_i} \in N_{0,1}(v)$. Thus $d_{p_0 q_0} * d_{q_1 q_1} \in N_{1,2}(v)$, and inductively $d_{p_0 q_0} * \dots * d_{p_t q_t} \in N_{t,t+1}(v)$. Thus

$$d = d_{p_0 q_0} * \dots * d_{p_t q_t} * \delta_x \in N_{t,k}(v).$$

Furthermore the foundation of d is the prescribed $k + t + 1$ subset u . We will call d a t, k -pod; δ_x where $|x| = k$ is called a $-1, k$ -pod. We should note that the multiplicities of a t, k -pod are always zero or plus or minus one.

In summary, the extension of an element of $N_{tk}(v)$ by a subset x disjoint from its foundation yields an element in $N_{t,k+|x|}(v)$ and the suspension of an element of $N_{tk}(v)$ by a pair of points disjoint from its foundation yields an element in $N_{t+1,k+1}(v)$.

4. MODULE BASES FOR $N_{tk}(v)$

In this section we will show that the t, k -pods generate $N_{tk}(v)$, and we will demonstrate techniques for the inductive construction of a module basis for $N_{tk}(v)$. Note that the t, k -pods are defined whenever $N_{tk}(v) \neq 0$.

4.1. Let $W = \{1, 2, \dots, v - 1\}$ and $Y = \{x \mid x \subseteq W\}$ where $v > 1$. We define $\varphi: Z^X \rightarrow Z^Y$ by:

$$(\varphi c)(x) = c(x + v), \quad \text{where } x \in Y \text{ and } v \text{ denotes } \{v\}. \quad (4.1)$$

Clearly φ is a linear transformation. One of the more useful properties of φ is that it "commutes" with G :

$$G(v - 1)\varphi = \varphi G(v). \quad (4.2)$$

We may see this directly by computing $\widehat{\varphi c}(u)$ for $u \in Y$:

$$\begin{aligned} \widehat{\varphi c}(u) &= \sum_{x \in Y} [u, x](\varphi c)(x) = \sum_{x \in Y} [u, x] c(x + v) \\ &= \sum_{x \in X} [u + v, x] c(x) = (\varphi \widehat{c})(u). \end{aligned}$$

A direct consequence of (4.1) and (4.2) is that, when $1 \leq t \leq k$, φ maps $C_{tk}(v)$ into $C_{t-1,k-1}(v-1)$ and $N_{tk}(v)$ into $N_{t-1,k-1}(v-1)$. Clearly (4.1) implies that, if c has constant block size k , then φc has constant block size $k-1$. Using (4.2) we get that, if c is a $(v, k, \lambda_0, \dots, \lambda_t)$ -design and $|u| = s < t$, then $\widehat{\varphi c}(u) = (\varphi \widehat{c})(u) = \widehat{c}(u+v) = \lambda_{s+1}$.

This operator also behaves well with respect to extensions and suspensions. One may easily verify that:

$$\varphi(\delta_v * c) = c, \quad \text{if } \text{found}(c) \subseteq W, \quad (4.3)$$

$$\varphi(d_{vq} * c) = c, \quad \text{if } v \neq q, \quad q \notin \text{found}(c) \subseteq W. \quad (4.4)$$

4.2. THEOREM. *For $0 \leq t$, $k \leq v$ and $v \geq 1$, either $N_{tk}(v) = \{0\}$ or $N_{tk}(v)$ has a module basis consisting of designs with foundation size $k+t+1$; in fact it has a module basis consisting of t, k -pods.*

We will prove this by induction on v . Unless $0 \leq t < k < v-t$, $N_{tk}(v) = \{0\}$; in particular the theorem is valid for $v=1$. Hence we may assume that $v > 1$, $t < k < v-t$. We would also like to assume that $N_{t,k}(v-1)$ and $N_{t-1,k-1}(v-1)$ have the required bases. With the exception of the case $t=0$, we may apply the induction hypothesis to these modules. In the case $t=0$, $N_{t-1,k-1}(v-1) = N_{-1,k-1}(v-1)$ and $\{\delta_x \mid |x| = k-1\}$ is the required basis.

Therefore we may assume $b_1, \dots, b_m$ to be a module basis for $N_{t,k}(v-1)$ where each b_i has foundation size $k+t+1$, and we may assume $c_1, \dots, c_n$ to be a module basis for $N_{t-1,k-1}(v-1)$ where each c_j has foundation size $k+t-1$ (of course n or m could be zero). Since $v-1 > k+t-1$, we may find $q_i \in W$, $q_i \notin \text{found } c_i$ for $i=1, \dots, n$. Then, by the results in paragraph 3.4, $d_i = c_i * d_{vq_i} \in N_{tk}(v)$.

Now let $c \in N_{tk}(v)$. We have noted that $\varphi c \in N_{t-1,k-1}(v-1)$, hence $\varphi c = \sum_{i=1}^n \gamma_i c_i$. Now consider $d = c - \sum_{i=1}^n \gamma_i d_i \in N_{tk}(v)$. We have:

$$\varphi d = \varphi c - \sum_{i=1}^n \gamma_i \varphi d_i = \varphi c - \sum_{i=1}^n \gamma_i c_i = 0.$$

It follows that $\text{found } d \subseteq W$ and hence $d \in N_{t,k}(v-1)$. This yields

$$d = \sum_{j=1}^m \beta_j b_j \quad \text{and} \quad c = \sum_{j=1}^m \beta_j b_j + \sum_{i=1}^n \gamma_i d_i.$$

We have then that $B = \{b_1, \dots, b_m, d_1, \dots, d_n\}$ generate $N_{tk}(v)$. However if $c=0$ above, we see that $\varphi c=0$. Since $\varphi b_j=0$, all of the γ_i are zero. It follows then that $d=0$ and the β_i are all zero. Thus B is a module

basis. Finally if the b_j were t, k -pods and the c_i were $t-1, k-1$ -pods then B would clearly consist of t, k -pods.

COROLLARY 1. *If $0 \leq t < k < v-t$, then $\varphi: N_{tk}(v) \rightarrow N_{t-1, k-1}(v-1)$ is onto.*

Since $N_{t-1, k-1}(v-1)$ has a basis with foundation size $k+t-1$, this basis may be suspended, as above, to elements in $N_{tk}(v)$ which then map onto the basis elements of $N_{t+1, k-1}(v-1)$ under φ .

COROLLARY 2. *If $0 \leq t < k < v-t$, then*

$$N_{tk}(v-1) \oplus N_{t-1, k-1}(v-1) \cong N_{tk}(v).$$

This is clear from the construction of the module basis.

4.3. As an application of the special module basis we give a theorem which is essential in proving the main result:

THEOREM. *Let $0 \leq t < k < v-t$, then the image of $N_{t, k}$ under $G_{t+1, k}$ is $N_{t, t+1}$.*

In paragraph 1.3 we indicated how one may interpret $G_{t+1, k}$ as operating on N_{tk} by mapping $c \in N_{tk}$ onto the element in Z_+^X with $t+1$ -blocks $G_{t+1, k}c_k$ and all other blocks zero. We noted at that time that $G_{t+1, k}c_k \in N_{t, t+1}$; it remains to show that the mapping is onto.

Let $c \in N_{t, t+1}$ such that $|\text{found}(c)| = 2t+2$. Since $v \geq k+t+1$, we may choose a $k-t-1$ subset x disjoint from $\text{found}(c)$. We have then that $\delta_x * c \in N_{tk}(v)$. If we can show that $G_{t+1, k}(\delta_x * c) = c_{t+1}$, we may conclude that $G_{t+1, k}$ is onto, since the elements with foundation size $2t+2$ generate $N_{t, t+1}$.

Now

$$G_{t+1, k}(\delta_x * c) = (\widehat{\delta_x * c})_{t+1}.$$

If $|y| = t+1$ we have from (3.2) that

$$\widehat{\delta_x * c}(y) = \begin{cases} 0, & \text{if } y \not\subseteq x \cup u, \\ \widehat{\delta_x}(xy) \mathcal{E}(uy), & \text{if } y \subseteq x \cup u, \end{cases}$$

$$\text{where } u = \text{found}(c).$$

We note that $\widehat{\delta_x}(xy) = 1$ for all y and

$$\mathcal{E}(uy) = \begin{cases} \mathcal{E}(y), & \text{if } y \subseteq u, \\ 0, & \text{if } |uy| \leq t. \end{cases}$$

Thus $(\widehat{\delta_x * c})(y) = c(y)$ if $|y| = t+1$, and $G_{t+1, k}(\delta_x * c) = c_{t+1}$.

5. THE EXISTENCE OF INTEGRAL $(v, k, \lambda_0, \dots, \lambda_t)$ -DESIGNS

MAIN THEOREM. *Let $t, v, k, \lambda_0, \dots, \lambda_t$ be integers where $v \geq 1$ and $0 \leq t, k \leq v$. There exists an integral $(v, k, \lambda_0, \dots, \lambda_t)$ -design if and only if*

$$\lambda_{s+1} = \frac{k-s}{v-s} \lambda_s \quad \text{for } 0 \leq s < t.$$

5.1. The necessity of the conditions for $t \leq k$ was proved in paragraph 1.2. If $t > k$, it follows directly from the definitions that $\lambda_s = 0$ for $k < s \leq t$.

We will prove the sufficiency of these conditions by induction on t . If $t = 0$ the set of conditions is vacuous and $\lambda_0 \delta_x$ is a (v, k, λ_0) -design, $|x| = k$. Assume then that the conditions are sufficient for some $t \geq 0$, and assume that $\lambda_0, \dots, \lambda_{t+1}$ satisfy the conditions. Then there exists an integral $(v, k, \lambda_0, \dots, \lambda_t)$ design c' . We would like to alter c' so as to make it an integral $(v, k, \lambda_0, \dots, \lambda_{t+1})$ design.

If k is outside the range $t < k < v - t$, we know that $C_{tk}(v)$ is one dimensional-spanned by e_k . In this case $c' = \alpha e_k$, and hence is a $(v, k, \lambda_0, \dots, \lambda_t, \lambda'_{t+1})$ -design we need only check that $\lambda_{t+1} = \lambda'_{t+1}$. This is a straightforward computation.

Now we turn to the case $t < k < v - t$. We have:

$$G_{tk} c_k' = \lambda_t e_t. \quad (5.1)$$

From section one we have

$$G_{tk} = \frac{1}{k-t} G_{t,t+1} G_{t+1,k},$$

and one may easily compute that

$$e_t = \frac{1}{v-t} G_{t,t+1} e_{t+1}.$$

Substituting these into (5.1) yields:

$$G_{t,t+1} G_{t+1,k} c_k' = G_{t,t+1} \frac{k-t}{v-t} \lambda_t e_{t+1} = G_{t,t+1} \lambda_{t+1} e_{t+1}.$$

If we let d' be the extension by zero of $G_{t+1,k} c_k' - \lambda_{t+1} e_{t+1}$, it is clear that $d' \in N_{t,t+1}$. By the theorem in paragraph 4.3, there exists $d \in N_{t,k}$ so that $G_{t+1,k} d_k = d'_{t+1}$. Finally, if $c = c' - d$ then c has constant block size k and

$$G_{t+1,k} c_k = G_{t+1,k} c_k' - d'_{t+1} = \lambda_{t+1} e_{t+1}.$$

It follows from Section 1 that the other parameters of c are $\lambda_0, \dots, \lambda_t$.

5.2. The mapping which assigns to each integral $(v, k, \lambda_0, \dots, \lambda_t)$ -design the integer λ_t is clearly a module homomorphism from $C_{t,k}(v)$ into Z . Therefore the image of this map is a submodule of Z . If $t > k$, this image is 0. If $t \leq k$, the image is not 0 and there exists a positive integer $l_t(v, k)$ which generates this submodule.

Since there exists an integral $(v, k, \lambda_0, \dots, \lambda_t)$ -design with $\lambda_t = l_t(v, k)$, we may compute λ_s to be

$$\frac{\binom{v-s}{t-s}}{\binom{k-s}{t-s}} l_t(v, k), \quad \text{for } s = 0, 1, \dots, t. \quad (5.2)$$

It follows from the main result that $l_t(v, k)$ is the smallest positive integer for which the numbers in (5.2) are all integral. Since

$$\frac{\binom{v-s}{t-s}}{\binom{k-s}{t-s}} = \frac{\binom{v-s}{t+1-s}}{\binom{k-s}{t+1-s}} \frac{k-t}{v-t},$$

we may prove inductively that $l_{t+1}(v, k)$ is the smallest positive multiple of

$$\frac{k-t}{v-t} l_t(v, k)$$

which is an integer (where $l_0(v, k) = 1$).

We conclude this paragraph by listing some of the basic properties of $l_t(v, k)$:

- (1) $l_t(v, k)$ divides $\binom{v-t}{k-t}$;
- (2) for $k = t$ or $k \geq v - t$, $l_t(v, k) = \binom{v-t}{k-t}$.

The first statement is obvious once we note that the design e_k has as its t -th parameter $\binom{v-t}{k-t}$, when $t \leq k$. Since e_k and its multiples are the only non-zero elements in $C_{t,k}(v)$, when $k \leq t$ or $k \geq v - t$, we conclude that the parameter of e_k is the appropriate parameter in these cases. This observation has a converse which we discuss in the next paragraph.

5.3. In this paragraph we gather together a few consequences of the main result which are formulated in terms of the $l_t(v, k)$.

COROLLARY 1. $G_{t,k}(v) c_k = \lambda_t e_t$ has an integral solution if and only if λ_t is a multiple of $l_t(v, k)$, where $t \leq k$.

COROLLARY 2. If $c \in C_{tk}(v)$ and $G_{tk}c_k = l_t(v, k) e_t$, then c and any module basis for $N_{tk}(v)$ form a module basis for $C_{tk}(v)$.

If $d \in C_{tk}(v)$ then $G_{tk}d_k = \lambda_t e_t$ and $\lambda_t = ml_t(v, k)$. Thus $d - mc \in N_{tk}(v)$.

Since we can effectively compute $l_t(v, k)$, the following corollary of the fundamental theorem is of interest:

COROLLARY. Let $t \leq k \leq v$. If $l_t(v, k) = \binom{v-t}{k-t}$, then the only t -design with block size k , which has as multiplicities only zeros and ones, is e_k .

If c is an element of $C_{tk}(v)$ which has as multiplicities only zeros and ones, then λ_0 is the number of k -blocks which have multiplicity one. Since λ_t is a multiple of $l_t(v, k) = \binom{v-t}{k-t}$ then λ_0 is a multiple of

$$\frac{\binom{v}{t}}{\binom{k}{t}} \binom{v-t}{k-t} = \binom{v}{k}.$$

It follows that $\lambda_0 = \binom{v}{k}$ and every k -block has multiplicity one, i.e., $c = e_k$.

We know of course that e_k is the only element of $C_{tk}(v)$ with multiplicities zero and one when $k \leq t$ or $k \geq v - t$. However $l_t(v, k)$ may equal $\binom{v-t}{k-t}$ within the range $t < k < v - t$. For example, $l_2(8, 3) = \binom{6}{1}$ as does $l_3(9, 4)$ and $l_4(10, 5)$.

REFERENCES

1. P. DEMBOWSKI, "Finite Geometries," Springer-Verlag, New York, 1968.
2. J. E. GRAVER AND W. JURKAT, Algebra structures for general designs, *J. Algebra* **23** (1972), 574-589.
3. W. B. JURKAT AND H. J. RYSER, Matrix factorization of determinants and permanents, *J. Algebra* **3** (1966), 1-27.